Open Quantum-State Control with Control-Dependent Decoherence

Cyril Morluyan Cordor[1], Anthony Bloch[1], Eitan Geva[2]

[1]Department of Mathematics, [2]Department of Chemistry, University of Michigan

Project Statement and Motivation

We investigate the controllability of open quantum dynamics as modeled by the **secular approximation to the Redfield quantum master equation (QME)** with control-dependent decoherence and compare dynamics modeled by the **Lindblad QME** with constant, control-independent dissipation.

- **Motivation.** Myriad new quantum technologies—including, among many, *nuclear magnetic resonance and medical spectroscopy*, *quantum sensing*, and *quantum computing*—drive the need to construct the proper framework modeling dynamics of open quantum systems (OQS). In real scenarios, quantum systems are never closed from the environment, and hence OQS dynamics are not unitary.
- **Challenge.** Entanglement of the system with the typically large number of environmental degrees of freedom causes **quantum decoherence**, the total non-unitary contributions to the dynamics [1]. Loss of coherence erodes the quantum properties underpinning the technology's advantages.

Control of Open Quantum Systems

Open Systems. An open quantum system S coupled with its environment/bath B is represented by a *Hermitian*, *positive-semidefinite*, density operator $\hat{\rho} \in \mathcal{L}(\mathbf{H})$ with *unity trace* over a Hilbert space $\mathbf{H} = \mathbf{H}_S \otimes \mathbf{H}_B$. For a 2-level system (2LS) (or a qubit), a system density operator $\hat{\rho}_S \in \mathcal{L}(\mathbf{H}_S) \cong \mathbb{C}^{2\times 2}$ and system Hamiltonian $\hat{H}_S$ can be written as

$$\hat{\rho}_S(t) = \frac{1}{2}I_2 + \frac{1}{2}\sum_{j=1}^{3} n_j(t)\hat{\sigma}_j, \quad \hat{H}_S(t) = \frac{\hbar}{2}\sum_{j=1}^{3} u_j(t)\hat{\sigma}_j, \qquad n_j(t),\, u_j(t) \in \mathbb{R}, \qquad (\mathcal{E}1)$$

where $\{\hat{\sigma}_j\}_j$ denote the Pauli matrices and $\{u_j\}_j$ are coherent controls. The 2 Hamiltonian energy levels are time-dependent, $E_{\pm}(t) = \pm\frac{\hbar}{2}\sqrt{u_1(t)^2 + u_2(t)^2 + u_3(t)^2}$.

OQS Control. Let $\mathcal{D}(\mathbf{H}_S) \subset \mathcal{L}(\mathbf{H}_S)$ be the set of admissible, system density operators. An open quantum system is **controllable** if there exists admissible controls $\{u_j\}_j$ that can steer a state $\hat{\rho}_S$ to any other arbitrary state in $\mathcal{D}(\mathbf{H}_S)$. *Complete* controllability of open systems is generally very difficult to obtain.

Markovian Quantum Master Equations (QME)

The true dynamics of an OQS is non-Markovian, but the Markovian approximations below [3, 4, 5] are valid for weak system-bath coupling and when the bath fluctuations timescale is much shorter than the system relaxation timescale, $\tau_c \ll \tau_R$.

Lindblad QME

$$\frac{d}{dt}\hat{\rho}(t) = -\frac{i}{\hbar}[\hat{H}(t), \hat{\rho}(t)] + \sum_{j=1}^{N}\gamma_j\left(\hat{L}_j\hat{\rho}(t)\hat{L}_j^{\dagger} - \frac{1}{2}\{\hat{L}_j^{\dagger}\hat{L}_j, \hat{\rho}(t)\}\right), \qquad (\mathcal{E}2)$$

$$N \leq d = \dim(\mathbf{H}_S)$$

- Lindblad/noise operators $\{\hat{L}_j\}_j$ and damping rates $\gamma_j > 0$ encode decoherence
- axiomatic derivation $\Longrightarrow$ no built-in prescription for each γ_j, $\hat{L}_j$

Secular Approximation to the Redfield QME

$$\frac{d}{dt}\rho_{jk}(t) = -i\omega_{jk}\rho_{jk}(t) - \frac{1}{2\hbar^2}\sum_{\ell=1}^{d}\rho_{jk}(t)\big(\check{\kappa}_{j\ell,\ell j}(\omega_{j\ell}) + \check{\kappa}_{k\ell,\ell k}(-\omega_{\ell k})\big) \qquad (\mathcal{E}3)$$
$$+ \frac{\delta_{jk}}{\hbar^2}\sum_{\ell=1}^{d}\rho_{\ell\ell}(t)\check{\kappa}_{\ell j,j\ell}(\omega_{\ell j}), \quad j,k \in \{1,\dots,d\}$$

- $\check{\kappa}_{jk,\ell m}$: Fourier transform of free-bath correlation functions $\kappa_{jk,\ell m}$ encode decoherence information directly from the physical attributes of the bath.

Bloch Vector Representation $\mathbf{n} \in \mathbb{B}$ for 2-Level System (2LS)

When $\dim(\mathbf{H}_S) = 2$, there's a bijective correspondence between the set of density operators $\mathcal{D}(\mathbf{H}_S)$ and the closed unit ball $\mathbb{B} \subset \mathbb{R}^3$ by taking the "traceless" components of $(\mathcal{E}1)$. (Dissipation here is pure dephasing and we set $\hbar = 1$.)

Lindblad-Bloch QME *with time- and control-independent dissipation*

$$\frac{d}{dt}\mathbf{n}(t) = \underbrace{\mathbf{h}(t) \times \mathbf{n}(t)}_{\text{coherent}} + \underbrace{(A - \mathrm{Tr}[A]I_3)\mathbf{n}(t) + \mathbf{b}}_{\text{Lindblad dissipation}}, \quad A \in \mathbb{R}^{3\times 3},\ \mathbf{b} \in \mathbb{R}^3 \qquad (\mathcal{E}4)$$

Redfield-Bloch QME *with control-dependent dissipation*

$$\frac{d}{dt}\mathbf{n}(t) = \underbrace{\Omega(t)\begin{pmatrix} n_2 \\ -n_1 \\ 0 \end{pmatrix}}_{\text{coherent}} - \underbrace{\frac{1}{2}K_a(t)\begin{pmatrix} n_1 \\ n_2 \\ 2n_3 \end{pmatrix} - D(t)\begin{pmatrix} n_1 \\ n_2 \\ 0 \end{pmatrix} + K_m(t)\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}_{\text{control-dependent dissipation of Lindblad-type}} \qquad (\mathcal{E}5)$$

$$\Omega(t) := \sqrt{u_1(t)^2 + u_2(t)^2 + u_3(t)^2}$$
$$K_{\substack{a \\ m}}(t) := (1 \pm e^{-\beta\Omega(t)})\frac{u_1(t)^2 + u_2(t)^2}{\Omega(t)^2}\check{\kappa}_{+-,-+}(\Omega(t)) \qquad (\mathcal{E}6)$$
$$D(t) := \frac{u_3(t)^2}{\Omega(t)^2}\check{\kappa}_{--,--}(0)$$

- The Redfield QMEs $(\mathcal{E}3)$, $(\mathcal{E}5)$ guarantee that the steady-state solution is the thermal equilibrium state, *i.e.* the Gibbs state $\hat{\rho}_S^{\text{eq}} = e^{-\beta\hat{H}_S} / \mathrm{Tr}[e^{-\beta\hat{H}_S}]$, where $\beta = 1/(k_B T)$ is the inverse thermodynamic temperature [2, 5].

Trajectories under Constant (Lindblad) vs. Control-Dependent (Redfield) Dissipation (Pure Dephasing Only)

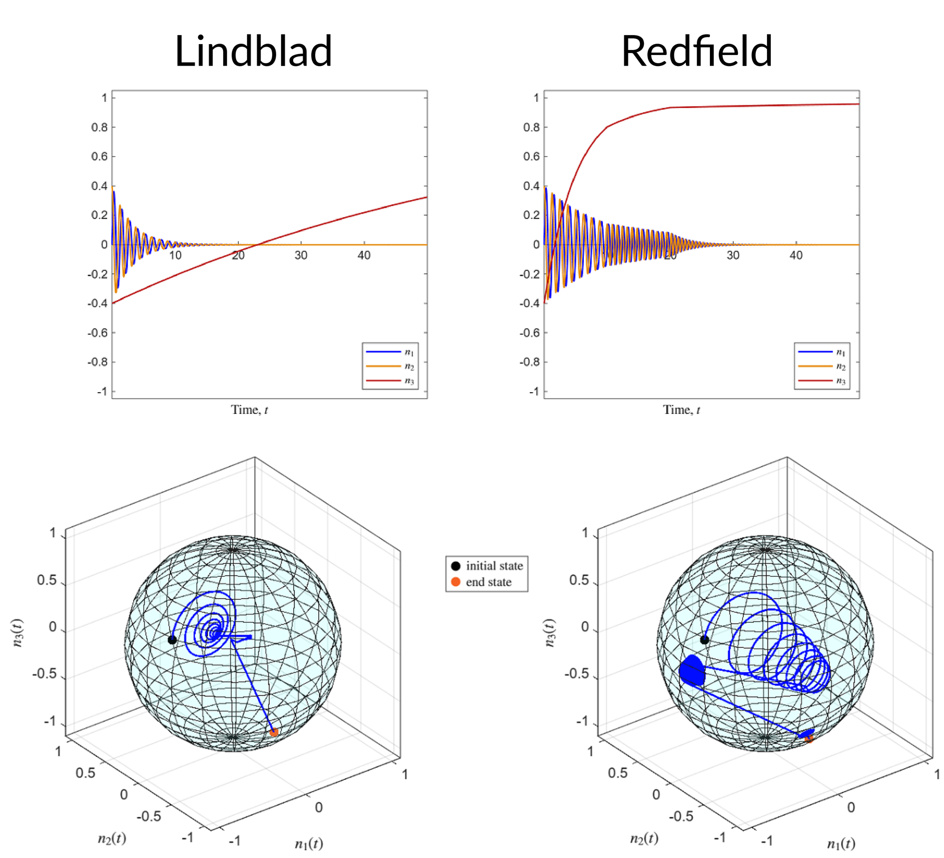

Figure 1: *Bloch trajectories steered by piecewise constant controls. Top row: n_1, n_2, n_3 components vs. time. Bottom row: 3D trajectories (blue) in $\mathbb{B}$ (light teal).*

- Bloch components n_1, n_2 store coherence/phase information, and n_3 information on ground/excited state populations.
- Under similar parameters, **Redfield** trajectories tend to capture coherence/phase behavior (precessions) in more detail than **Lindblad**.
- **Redfield** shows that with a non-stationary Hamiltonian $\hat{H}_S(t)$, pure dephasing also converts to population relaxation and energy dissipation.

References

[1] H.P. Breuer and F. Petruccione. *The Theory of Open Quantum Systems*. Oxford University Press, New York, 2002.

[2] Roie Dann and Ronnie Kosloff. Open system dynamics from thermodynamic compatibility. *Physical Review Research*, 3(2):023006, May 2021.

[3] V. Gorini, A. Kossakowski, and E.C.G. Sudarshan. Completely positive dynamical semigroups of n-level systems. *Journal of Mathematical Physics*, 17(5):821–825, May 1976.

[4] G. Lindblad. On the generators of quantum dynamical semigroups. *Communications in Mathematical Physics*, 48(2):119–130, June 1976.

[5] A.G. Redfield. On the theory of relaxation processes. *IBM Journal of Research and Development*, 1(1):19–31, 1957.

[6] P. Rooney, A. Bloch, and C. Rangan. Flag-based control of quantum purity for $n = 2$ systems. *Physical Review A*, 93(6):063424, June 2016.

The Purity Problem

Definition. The **purity** of a quantum state $\hat{\rho} \in \mathcal{L}(\mathbf{H})$ is given by $\mathrm{Tr}[\hat{\rho}^2]$. Purity measures how close a quantum state is to an isolated system or *pure state* which has purity $\mathrm{Tr}[\hat{\rho}^2] = 1$.

- For $\mathrm{Tr}[\hat{\rho}^2] < 1$, $\hat{\rho}$ is a *mixed state*; the maximally mixed state has purity $= 1/\dim(\mathbf{H})$ and represented by $\mathbf{0} \in \mathbb{B}$ for a 2LS. The less the purity, the greater the uncertainty in information of the state of the system.
- Coherent controls can only steer along unitary orbits (density operators with same spectrum) which have constant purity. Hence, **the irony is that decoherence is needed to steer the state among different purity levels.**
- For a 2LS, the purity $\mathrm{Tr}[\hat{\rho}^2] = \frac{1}{2}(1 + \|\mathbf{n}\|^2)$. Thus, unitary orbits are one-to-one with concentric spheres inside $\mathbb{B}$.

Radial-Transverse Decomposition of Bloch QMEs

Let $\mathbf{n} = r\hat{\mathbf{n}} \in \mathbb{B}$ where $r = \|\mathbf{n}\| \in [0, 1]$ and $\hat{\mathbf{n}}$ the Bloch unit vector. The Bloch QMEs can be decomposed into radial and transverse components,

$$\frac{dr}{dt} = \mathbf{b}\cdot\hat{\mathbf{n}} + r(\hat{\mathbf{n}}\cdot A\hat{\mathbf{n}} - \mathrm{Tr}[A]), \qquad \frac{d\hat{\mathbf{n}}}{dt} = \mathbf{h}\times\hat{\mathbf{n}} + (A\hat{\mathbf{n}})_{\perp} + \frac{1}{r}\mathbf{b}_{\perp}, \qquad (\mathcal{E}7)$$

where $\perp$ denotes the component perpendicular to $\hat{\mathbf{n}}$. We call the radial component the **purity derivative** [6].

- The radial component of **Lindblad-Bloch** $(\mathcal{E}4)$ has no explicit dependence on the Hamiltonian control vector $\mathbf{h} = (u_1, u_2, u_3)$. One must control $\hat{\mathbf{n}}$ to influence the purity derivative [6].
- The purity derivative of **Redfield-Bloch** $(\mathcal{E}5)$ has explicit dependence on the Hamiltonian, via $K_{a,m}(t)$, $D(t)$ $(\mathcal{E}6)$, due to control-dependent dissipation,

$$\frac{dr}{dt} = K_m(t)\hat{n}_3 - r\left[D(t)(\hat{n}_1^2 + \hat{n}_2^2) + \frac{1}{2}K_a(t)\hat{n}_3^2\right]. \qquad (\mathcal{E}8)$$

Implication. The Redfield-Bloch equation $(\mathcal{E}5)$ demonstrates that Hamiltonian controls have more direct influence on purity than predicted by Lindblad with constant, control-independent dissipation.

Controlling Bloch Vector $\mathbf{n} \in \mathbb{B}$ to Purity-Rising Regions

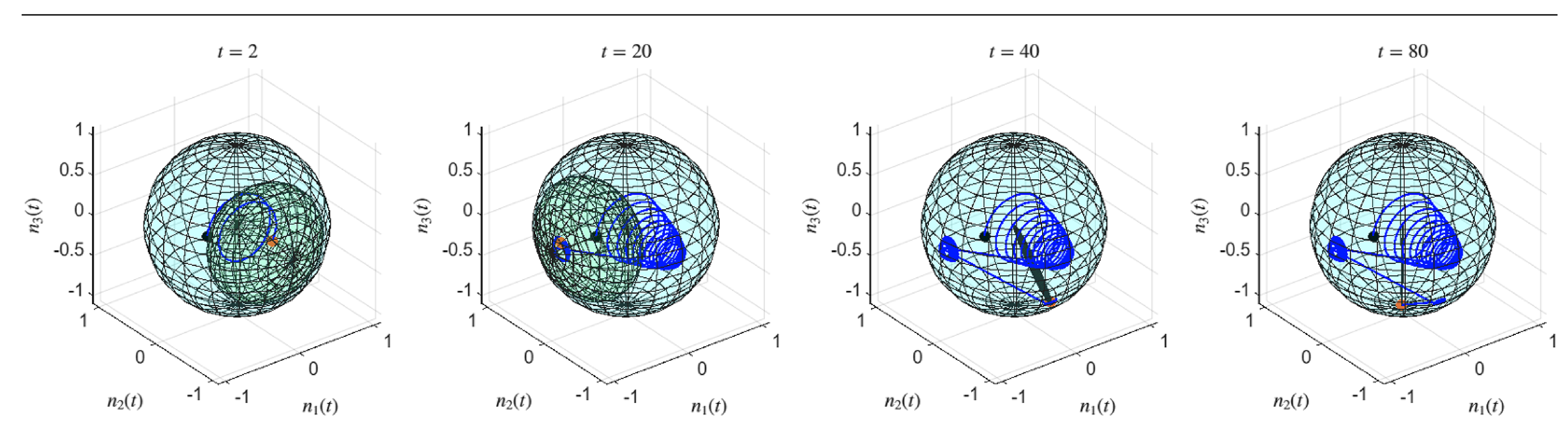

Figure 2: *Redfield-Bloch trajectory at varying times plotted with purity-rising chimneys (light green).*

- One control objective might be to maximize the purity (or minimize uncertainty) of a quantum state $\hat{\rho}_S$, which means steering $\mathbf{n} \in \mathbb{B}$ towards the boundary $\partial\mathbb{B} = \mathbb{S}^2$.
- The zero level set of the purity derivative in $(\mathcal{E}7)$ forms an ellipsoid. In its interior, which we call the **chimney**, purity rises since $dr/dt > 0$; the point furthest from the origin is the **apogee** [6].
- For **Redfield-Bloch** $(\mathcal{E}5)$ trajectories: (1) Thermal equilibrium $\hat{\rho}_S^{\text{eq}} \leftrightarrow \mathbf{n}_{\text{eq}}$ *is the* apogee. (2) Simple coherent controls can be constructed to naturally steer trajectories into the chimney as well as achieve long-term stability even when the chimney shrinks.

Github: https://github.com/cyril-cordor/oqs-sec-redfield-lindblad-control/

ccordor@umich.edu